



**Brecksville-
Broadview Heights
City School District**

SUSTAINABILITY

April 24, 2024

Brecksville - Broadview Heights City School District

What is sustainability?

Sustainability is defined as meeting the needs of the present without compromising the ability of future generations to meet their own needs.

Environmental sustainability involves managing resources and fostering practices that help protect natural systems, minimize pollution and environmental degradation, and maintain the health of the planet's ecosystems.

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What are we doing?

Mechanical System - HVAC

- Building Automation
- Occupancy Schedules
- Economizing Outside Air
- VFD's on Pumps & Motors
- High Efficiency MS Boilers

Custodial Supplies

- Coreless Toilet Paper
- FSC Paper Towels
- Green Seal Chemicals
- EcoLogo Hand Soap



Plumbing Systems

- Bottle Filling Stations
- Low Flow Fixtures
- Automated Sinks & Flush Valves

Electrical - Lighting & LED

- ES All Lighting + Automation
- HS & MS Auditoriums
- HS Main & Aux Gym
- Board Room
- MS Supplemental Skylights
- External Photocells & Timers

Other

Classroom / Building Recycling
Added Roof Insulation
Energy Star Technology

Anti-Idling Policy
Stormwater Management
Low/No VOC (Adhesives/Paint)

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What can we improve? (low hanging fruit)

Recycling

- Classroom Paper Recycling Bins
 - \$20/bin (\$3,000 - \$4,000)
 - Possible Ohio EPA Grant
- Stakeholder Awareness
 - FALSE *"Put everything in the bin, the hauler will sort everything anyway"*
 - FALSE *"If it's made of plastic it can be recycled"*
 - FALSE *"recycling arrows mean it is recyclable"*
 - FALSE *"The paper only dumpsters are a money maker"*

Additional Bottle Filling Stations

- Ohio EPA Grant (October 2024)
- \$5,000-\$10,000

Facility Audits

- HVAC Retro Commissioning
- Trane Feasibility Study
- Ameresco Energy Assessment
- Campus Master Plan

Stakeholder Assistance

- Reduce, Reuse, Recycle
- Windows and Thermostats
- K-Cups / Disposable cups
- Water bottles
- Carpool / Walk / Bike
- Power off when not in use

What can we improve? (harder to obtain items)

LED Upgrades

- Exterior Lighting
 - Building Perimeter
 - Entryway
 - Walkway
 - Parking Lot
 - Stadium
- Interior Lighting
 - Cafeterias
 - Corridors
 - Classrooms
 - Restrooms
- Automation
 - Programing
 - Schedules

Mechanical Systems

- HS Boilers
- HS Chillers
- HS Pumps & Motors

Cafeteria

- Reusable Trays/Plates
- Enhance/Increase School Gardens
- Incorporate Composting

Potential Initiatives

- Student Driven Sustainability Project/Art
- Paper Limits
- Copy Count Limits
- Pesticide (weed/fertilizer) Reduction

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What's not so feasible? (at this time)

Thermal Batteries

- Utilizes ice for facility cooling
- Chiller with supplemental ice storage
- Chiller with full ice storage
- Infrastructure
- Increased Maintenance
- Increased Controls
- Return on Investment
- Ohio Weather

Electric Buses

- Infrastructure
- Technicians / Mechanics
- Life Expectancy
- Battery Disposal
- Ohio Weather

Facility Solar Power

- Orientation of Facilities
- Available / Suitable Roofs
- Infrastructure
- Life Expectancy
- Return on Investment
- Ohio Weather

A second look...

HS Chiller Replacement

- Energy Savings
- Better Refrigerant
- Potential IRA Tax Credits
- Maintenance Savings

Electric Vans

- Reduced Fuel Cost
- No Emissions
- Noise Reduction
- Educational Opportunities

Smaller Scale Solar

- Campus Master Plan
- Off Grid Lighting (away from building power)
 - Walkways
 - Additional Parking Lights
 - Additional Drive Lights
- Electric Vehicle Solar Charging Stations

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Where do we go from here?

District Sustainability Leaders

- Shirley Tichy
- Emily Koziol
- Morgan Kolis
- Cathy Roderick
- Sheryl Lewis
- Heather Luckey
- Mark Miller
- Ashley Anders

High School Biology Club

- Lou Rundo
- Students

Self Reflect

- What are you doing
- What can you do better
- Volunteer / Participate
- Plant a tree
- Pick up litter
- Reduce use
- Reuse items
- Recycle (the right way)
- Home compost
- Home garden
- Buy sustainable
- Reusable lunch containers
- Dress for the weather

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QUESTIONS & DISCUSSION

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